

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE**Supplementary Winter Examination – 2025****Course: B.Tech.****Branch: Common for all branches****Semester: II****Subject Code & Name: BTES205E ENERGY AND ENVIRONMENT ENGINEERING****Max Marks: 60****Date: 22/01/2026****Duration: 3 Hr.****Instructions to the Students:**

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

		(Level/CO)	Marks
Q. 1	Objective type questions. (Compulsory Question)		12
1	The main working fluid in a steam power plant is		1
	a. Air b. Steam c. Water d. Gas		
2	Biogas mainly contains		1
	a. Oxygen b. Hydrogen c. Methane d. Nitrogen		
3	Energy conservation means		1
	a. Stopping energy use b. Efficient use of energy c. Producing more energy d. Wasting less water		
4	AQI stands for		1
	a. Air Quality Index b. Air Quantity Index c. Air Quality Indicator d. Atmospheric Quality Index		
5	Noise pollution is measured in		1
	a. Watt b. Hertz c. Decibel d. Pascal		
6	Variable speed drive helps in saving energy in		1
	a. Pumps b. Fans c. Compressors d. All Above		
7	MHD power generation uses		1
	a. Solid fuel b. Liquid fuel c. Plasma d. Steam		
8	COP is related to		1
	a Boilers b. Turbines c.Refrigeration d. Generators		

9	Acid rain is mainly caused by					1
	a. CO ₂	b. SO ₂ and NO _x	c. CH ₄	d. O ₂		
10	Which turbine is used for low head hydro power plants?					1
	a. Pelton	b. Francis	c. Kaplan	d. Impulse		
11	Solar constant value is approximately					1
	a. 500 W/m ²	b. 1000 W/m ²	c. 1367 W/m ²	d. 2000 W/m ²		
12	Ozone layer depletion is mainly due to					1
	a. CO ₂	b. SO ₂	c. CFCs	d. NO ₂		
Q. 2	Solve the following.					12
A)	Derive efficiency of thermoelectric generator and list its limitations.					6
B)	Explain role of deaerator in thermal power plants.					6
Q.3	Solve the following.					12
A)	Explain variable speed drives and their role in energy conservation.					6
B)	Discuss methods for measurement of air pollution.					6
Q. 4	Solve Any Two of the following.					12
A)	Discuss significance of BOD and COD in water quality assessment.					6
B)	Explain thermal pollution and its effect on aquatic life.					6
C)	Explain role of insulation in energy conservation with examples.					6
Q.5	Solve Any Two of the following.					12
A)	Explain primary and secondary air pollutants with examples.					6
B)	Explain advanced wastewater treatment methods.					6
C)	What is gas turbine power plant?					6
Q. 6	Solve Any Two of the following.					12
A)	State advantages of LED lamps.					6
B)	What is soil pollution?					6
C)	What is MHD power generation?					6
	*** End ***					